MCT344 & MCT342

# Industrial Robotics

## Sheet (1): Part#2

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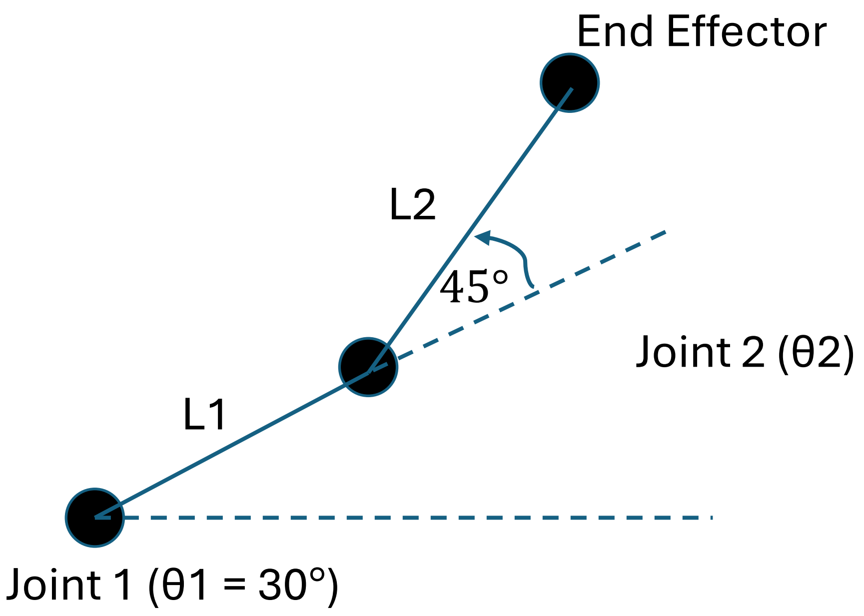
# Sheet (1) – Pb.23

A 2R planar arm has L1 = 4 cm, L2 = 3 cm. Joint angles are theta1 = 30 degrees and theta2 = 45 degrees (theta2 relative to link-1). Compute the end-effector position (x, y).



# Solution:

theta12 = theta1 + theta2 = 30 + 45 = 75 degrees.

x = L1*cos(theta1) + L2*cos(theta12),

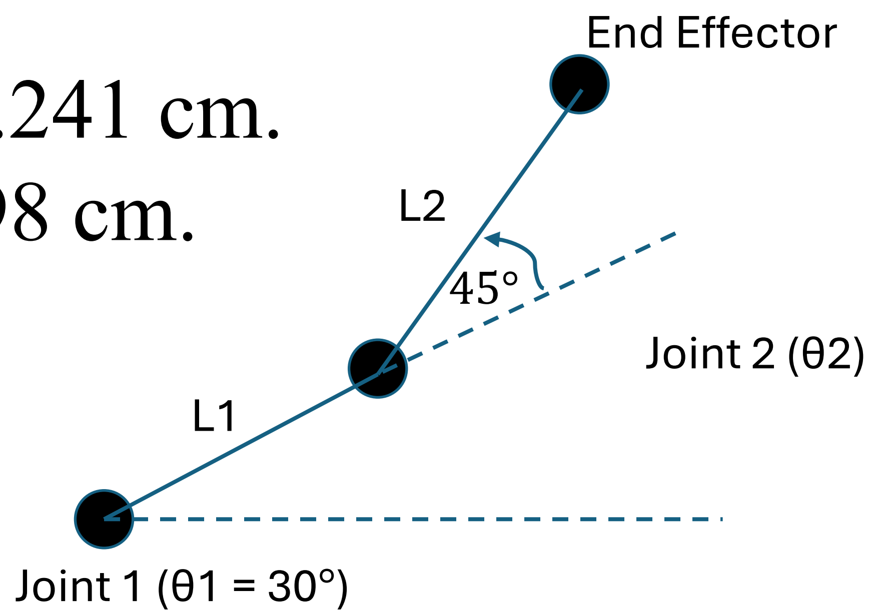
y = L1*sin(theta1) + L2*sin(theta12).

cos30=0.866, sin30=0.5

cos75=0.259, sin75=0.966.

x = 4*0.866 + 3*0.259 = 3.464 + 0.777 = 4.241 cm.

y = 4*0.5 + 3*0.966 = 2.000 + 2.898 = 4.898 cm.



# Sheet (1) – Pb.24

A 2R arm has L1 = 4 cm, L2 = 3 cm. Target is P = (3,4) cm.

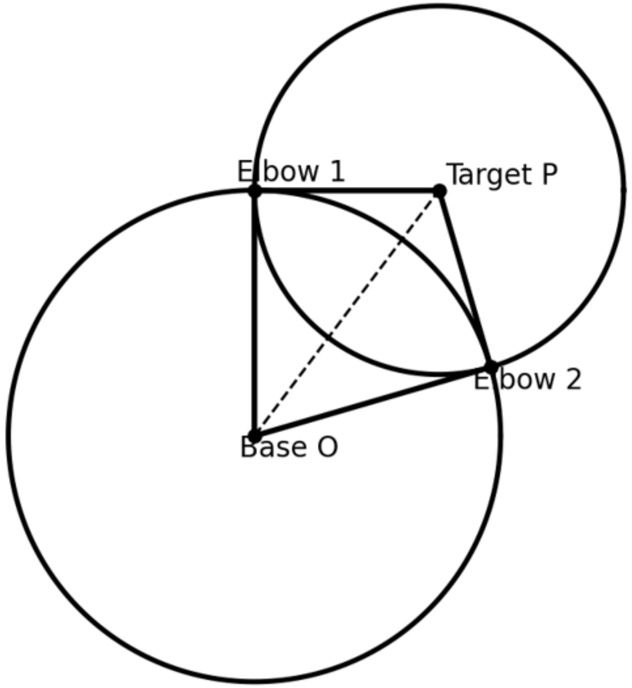
a) Using compass: draw a circle centered at O with radius L1 and a circle centered at P with a radius L2.

b) How many intersections are there? What does that mean for IK solutions?

c) Use cosine rule to compute the elbow angle theta2.

# Solution:

a) Draw circle(O, L1) and circle(P, L2).



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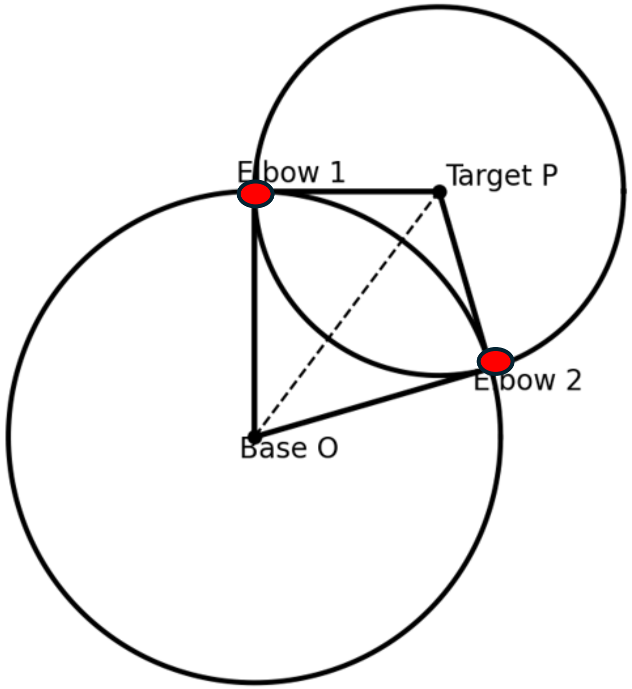
# Solution:

b) 2 intersections are existing.
therefore, there are two IK solutions.
(elbow up and elbow down)



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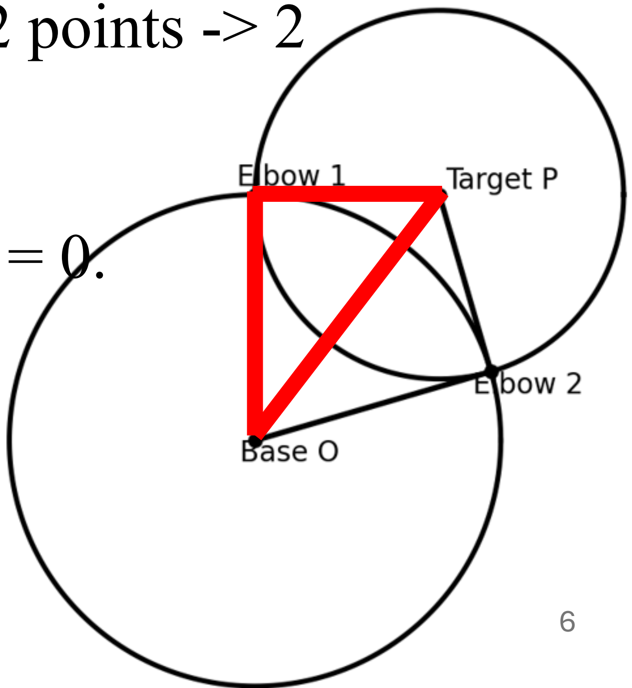
# Solution:

c) For these values, distance OP = 5 and the circles intersect in 2 points -> 2 solutions.
Using cosine rule on triangle (O, elbow-up, OP=5):
cos(theta2) = (L1^2+L2^2-d^2)/(2*L1*L2) = (16+9-25)/(24) = 0.
So, theta2 = 90 degrees.



# Grübler's Formula

$$DOF = m(N - J - 1) + \sum_{i=1}^{J} f_i - f_p$$

m: m = 6 for spatial bodies and m = 3 for planar mechanisms.
N: Number of bodies, including ground.
J: Number of joints.
$f_i$: Degrees of freedom for each joint.
$f_p$: Number of passive degrees of freedom.

| Joint Type | Degrees of freedom |
|---|---|
| Revolute | 1 |
| Prismatic | 1 |
| Helical | 1 |
| Universal | 2 |
| Cylindrical | 2 |
| Spherical | 3 |

## Sheet (1) – Pb.15

Consider a planar four-bar linkage consisting of four rigid links connected in a single closed loop by four revolute (pin) joints.

Using the **Gruebler–Kutzbach** mobility criterion, determine the degrees of freedom (mobility) of the mechanism.

Assume:

Motion is planar.

All joints are 1-DOF revolute joints.

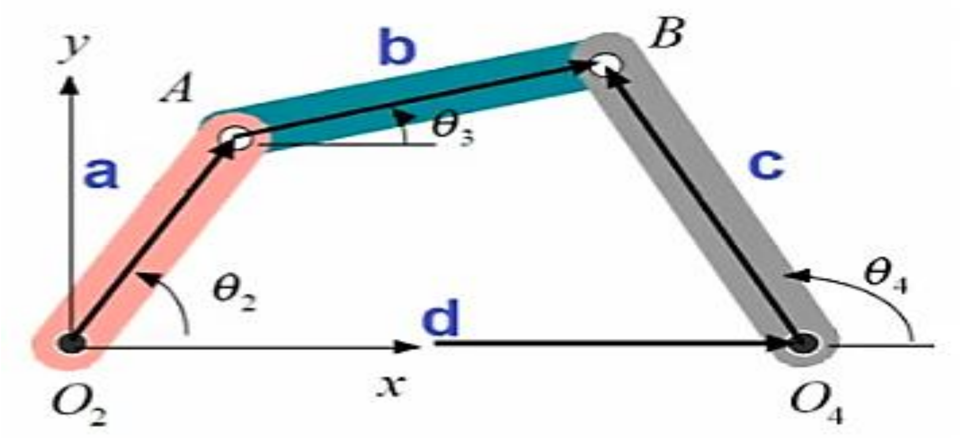



Fig.4 Four-bar mechanism

# Solution

$$DOF = m(N - J - 1) + \sum_{i=1}^{J} f_i - f_p$$

m = **3** for **planar** mechanisms.

N = 4

J = 4.

$f_i = 1.$

$f_p$ = 0

$DOF = 3(4 - 4 - 1) + $ 4x1 - 0 = -3 + 4 - 0 = 1

# Sheet (1) – Pb.16

- A Stewart–Gough platform shown in **Fig.3** consists of:
- A fixed base
- A moving platform
- 6 identical extensible legs connecting the base to the platform
- **Each leg has:**
- 1 Prismatic joint (P) actuated (changes length)
- 2 Spherical joints (S) one at the base, one at the platform
- So, each leg has a joint sequence:

$S$ - $P$ - $S$

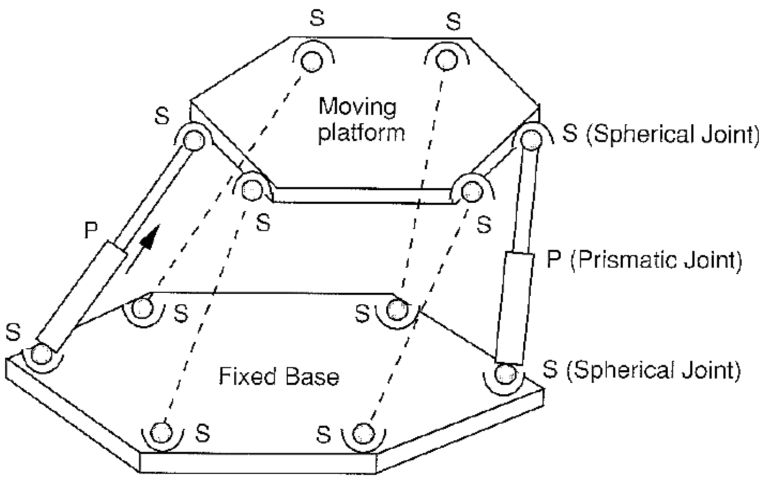



**Fig.3 Stewart–Gough platform**

Using mobility **Gruebler–Kutzbach** criterion for spatial mechanisms, determine:
What is the total number of degrees of freedom (DOF) of the moving platform relative to the base?

# Solution

$$DOF = m(N - J - 1) + \sum_{i=1}^{J} f_i - f_p$$

m = **6** for **spatial** mechanisms.

N = 2 + 6*2 = 14

J = 6 spherical + 6 Prismatic + 6 spherical = 18 .

$f_{spherical} = 3, f_{prismatic} = 1.$

$f_p$ = 6

$DOF = 6(14 - 18 - 1) +$ (12 x 3 + 6*1) $-$ 6= -30+ 42 - 6 = 6

# Sheet (1) – Pb.17

A 2R arm has L1 = 80 mm and L2 = 60 mm. Consider three targets P at distances from the base:

Case A: d = 140 mm

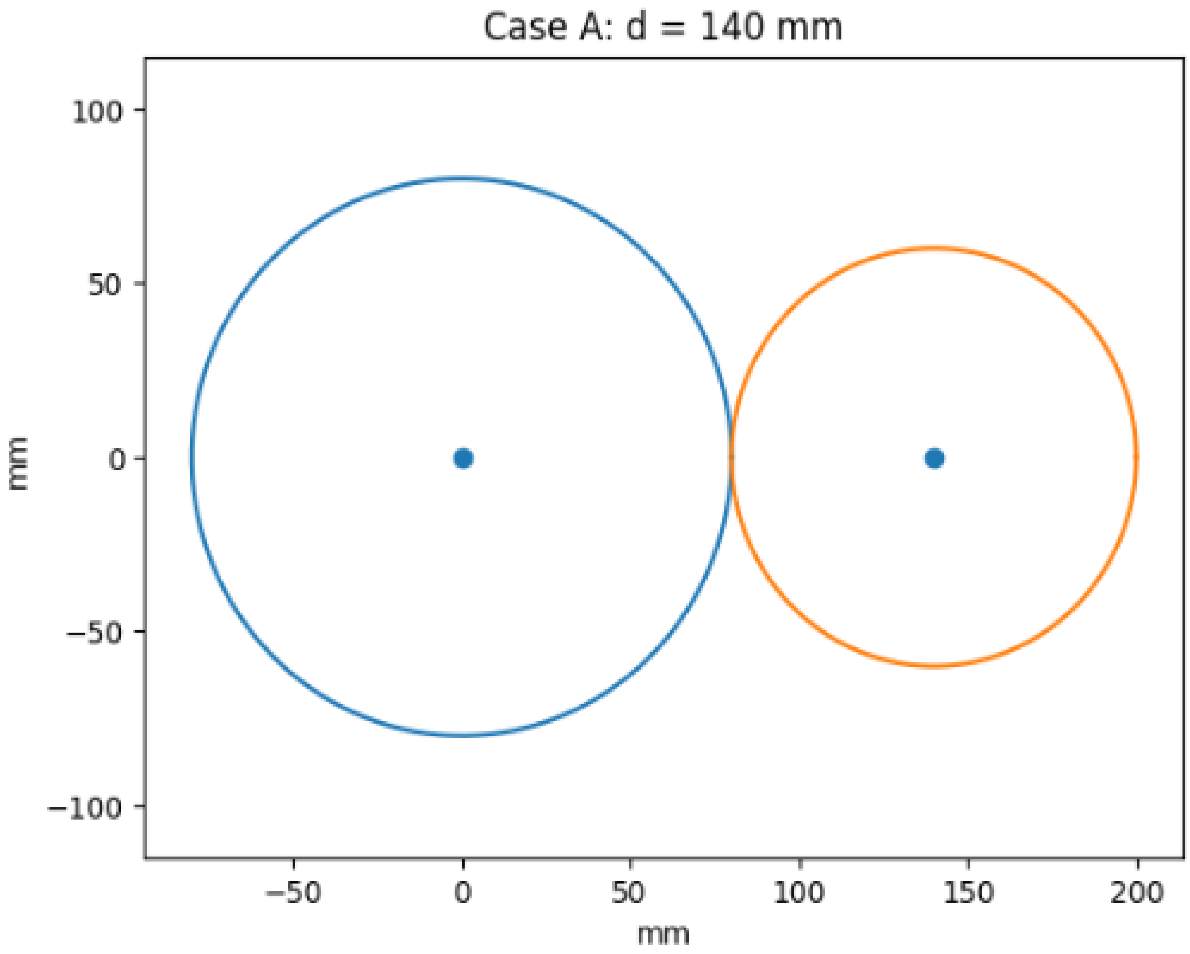
Case B: d = 100 mm

Case C: d = 10 mm

For each case, determine graphically (using circles) the number of IK solutions and whether the configuration is singular.

# Solution

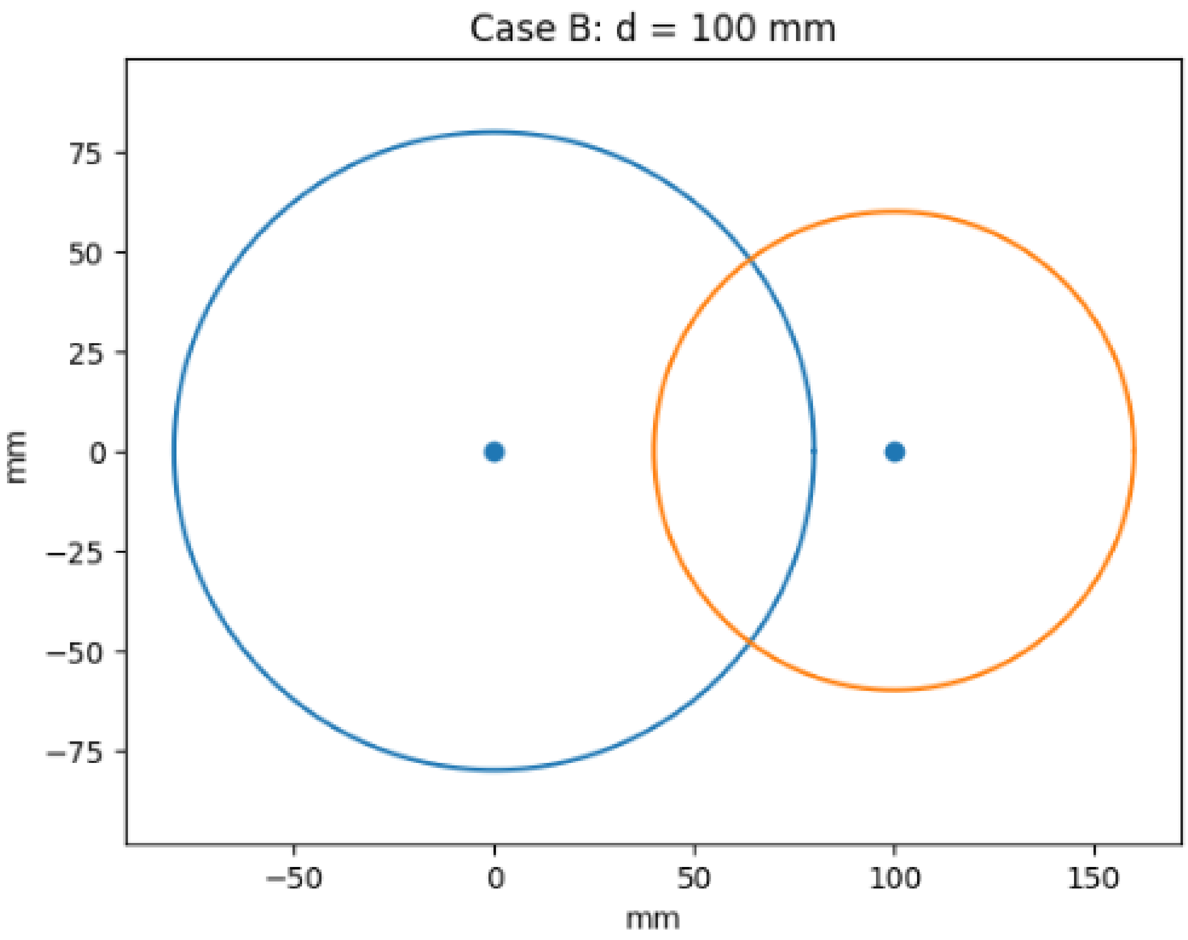
- **Maximum reach** = $L_1 + L_2$ = 140 mm
- **Minimum reach** = $|L_1 - L_2|$ = 20 mm




- **Number of IK solutions:** 1
- **Configuration:** Singular
- **Reason:** The elbow is fully extended → loss of one degree of freedom.

# Solution

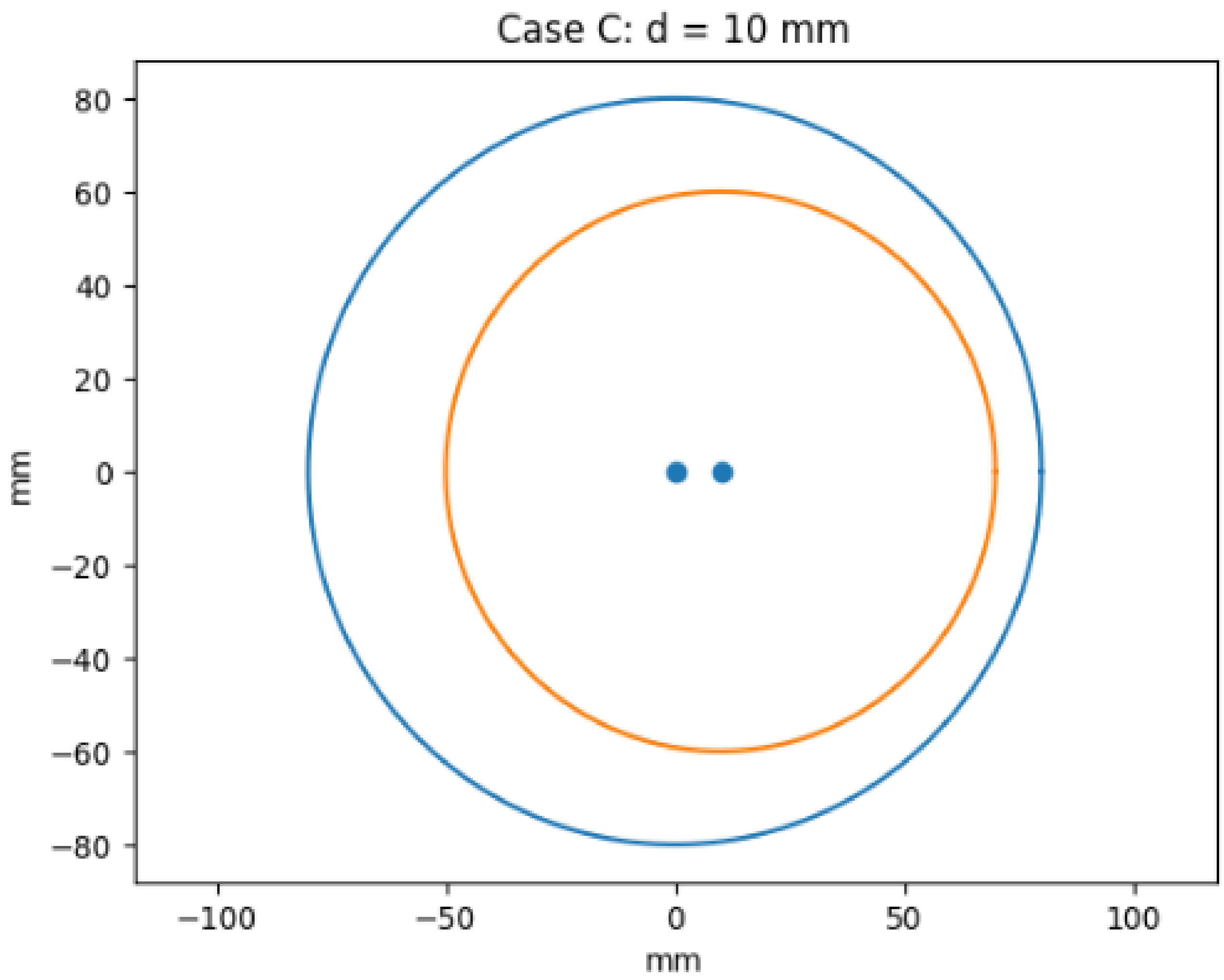
- **Maximum reach** = $L_1 + L_2$ = 140 mm
- **Minimum reach** = $|L_1 - L_2|$ = 20 mm




- **Number of IK solutions:** 2
- **Configuration:** Non-Singular
- **Reason:** The arm has two distinct postures.

# Solution

- **Maximum reach** = $L_1 + L_2$ = 140 mm
- **Minimum reach** = $|L_1 - L_2|$ = 20 mm




- **Number of IK solutions:** 0
- **Configuration:** No reachable solution
- **Reason:** The arm can't fold to reach that close.

# Any questions?